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# Exploring and Experimenting with Bundles of 'Hacked' Practices as Prototyping for Sustainable Transitions

**Maria Vitaller del Olmo, Amalia de Götzen, Nicola Morelli, Luca Simeone**

Within transition theory, practices have been identified as the focus unit for transforming established landscapes. Operating at different levels of everyday life and responding to needs, practices are embedded in our societies and contribute to existing sustainability challenges. While previous research focuses on the theoretical integration of social practice theory and transition theory, there is still a lack of studies focusing on the methodological implications for designers working within transition design processes on operating with practices. This paper examines the role of prototyping, a widespread design methodology, in the transformation of complex sociotechnical systems. More specifically, we focus on the methodological implications of prototyping existing and new practices that can contribute to establishing sustainable landscape transitions.

Following the Jerez de la Frontera case study in the PALIMPSEST EU-funded research project, we apply grounded theory-building techniques to examine 1) the methodological and practical application of prototyping to practices and 2) the influence of prototyping 'hacked' practices in design processes addressing sustainable landscape transitions. The Jerez team started from existing practices (e.g., ceramic crafting, traditional flamenco performances, vine growing). These practices, despite being strongly rooted in the local context, were unrelated among them, some even enacted outside the local landscape. Early prototyping allowed the team to deeply understand the elements and actors participating in the targeted practices, so they could envision a solution ecosystem 'hacking' such practices. That is, the practices were reconstructed either in relation to the socio-technical system – becoming more endogenous and/or bundled – or in their components – by altering the materials, meanings, and

competences associated with them. Then, the team prototyped the 'hacked' practices and evaluated their adoption, enactment and potential in relation to the envisioned landscape transformation. Our longitudinal study provides relevant contributions to transition design theory and practice. Theoretically, we describe the perceived difficulty in separating 'prototyping' and 'testing' activities, unfolding academic reflections on the redefinition of long-established design practices when applied to complex system transformations. Methodologically, we appreciated clear differences between the filtering and manifestation dimensions of earlier and later prototypes in this process that can guide and inspire the work of practitioners in similar projects. For example, our analysis indicates four categories of the prototyping activities according to their focus on aspects of the practices that had been edited – their elements, their enactment by local actants in the local landscape, their newly-established interrelation, and the necessary adjustment of norms for the practices to operate in the context. Another important methodological implication of our study refers to the predominance of high-fidelity prototypes to engage participants and trigger the envisioned transformation in socio-technical systems.

KEYWORDS: prototyping, practice prototyping, sustainable transitions

RSD TOPIC(S): Cases & Practice, Methods & Methodology, Sociotechnical Systems

## Introduction

Unsustainable practices, fostered by the idea of infinite growth (Meadows et al. 1972) are severely affecting urban and rural landscapes and their ecosystems (Jacobs 1961; Forman 2014). As a response, researchers, practitioners and decision-makers seek ways to transition towards more sustainable and resilient environments. Transition theory, drawing from social practice theory, appoints practices as the focus unit for transformation. Defined as "socially shared patterns of activity" (Watson and Shove 2022), practices are deeply rooted in societies' everyday life and contribute to existing sustainability challenges. Consequently, they may also contribute to mitigate or alleviate such challenges. Like Irwin et al. (2020) and Watson and Shove (2022), we pay special attention to practices as essential units for transformation processes that can conform

niches of innovation at any level of everyday life (Wallace 2021) and have, therefore, the potential to unfold transitions in the long term. Most of the available research focuses on the integration of social practice theory and transition theory from a theoretical perspective (e.g. Kossoff 2008/2011, Irwin et al. 2020); consequently, the actual application of practices in complex transformations remains underexplored. This paper aims to contribute to closing this gap by examining the role of prototyping, a long-established design methodology, in the transformation of complex socio-technical systems. Particularly, we focus on the methodological implications of prototyping practices to establishing sustainable landscape transformations.

Prototyping has proven its great potential in collaborative processes addressing the design and implementation of innovation products or services (Brown 2009, Ehn et al. 2014) and is recognised as a promising method to facilitate design processes in complex, uncertain and rapidly evolving systems (Bijl-Brouer and Malcolm 2020, Kimbell and Bailey 2017, Drew et al. 2021). However, the methodological specificities of the application of prototyping in transition design processes remain underexplored, and more research on the topic is required (e.g. Maiorana 2021, The Design Council 2021). In particular, we are interested in the practical details of prototyping new and existing practices due to their relevance in complex sociotechnical transition processes. To do so, we study the work developed by Jerez de la Frontera (Spain), one of the pilots of the European research project PALIMPSEST. The Jerez team prototyped a solution ecosystem consisting of an aggregation (bundle) of 'hacked' practices (e.g. teaching ceramic crafting from a bioregional and cultural approach, collective singing and dancing for provoking a dialogue on renewable energy, renaturalization of the urban landscape for climate mitigation). The team 'hacked' rooted practices by reconstructing their relation with the socio-technical systems in Jerez (becoming more endogenous and/or aggregated) and/or their components (altering the materials, meanings and competences associated with them). Contrary to more conventional design processes, the object of the Jerez prototypes were practices, as they constituted key leverage points in the solution ecosystem envisioned by the team to promote a sustainable landscape transformation. The prototypes developed by the team served to 1) understand the existing practices in the local contexts and their key elements and actants and 2) evaluate the adoption and impact of the proposed bundle of 'hacked'

practices to establish their envisioned transformation in the local landscape. The literature distinguishes two main prototyping dimensions: filtering and manifesting (Lim et al. 2008). The filtering dimension of prototyping is understood as the selection of aspects of a design idea, focusing on understanding, learning about, or validating particular elements of an envisioned design space. The manifesting dimension is defined as the "physical" externalization and realization, in the most effective and simplest form, of the selected aspects of the design idea to facilitate their exploration and evaluation. In our longitudinal study of the Jerez case, we collected data about the filtering and manifesting dimensions of the prototypes developed by the team for their envisioned landscape transformation, as well as information about the testing of such prototypes in the local context. This paper reports on our findings, which show clear differences in the filtering and manifestation dimensions of prototyping between earlier and later prototyping initiatives, as well as necessary theoretical reflections on the differentiation of prototyping and testing practices when applied to complex socio-technical transitions. The paper also provides methodological considerations and suggestions for the adoption of the prototyping practice in similar projects, attending to the insights obtained.

## Literature Review

Multi-Level Perspective, the Domains of Everyday Life in Transition Design and the concept of nested holism

The Multilevel Perspective (MLP) framework offers an analytical perspective to study change and transformation within socio-technical systems (networks of actors, artifacts, activities, institutions, rules and norms). According to the MLP, there are three levels of scale – niches, regimes and landscapes (Geels 2010, Geels and Schot 2007, 2010) – that are used to describe (and eventually influence) the dynamics within a socio-technical system and its transition potential (Rotmans and Loorbach 2010). The levels differ in their stability, that is, their resistance or propensity to change (Geels and Schot 2010). At the micro level, niches are more unstable and open to alterations and innovation. At the meso level, regimes present "more stable configurations" of regulations, markets, and artefacts. At the macro level, landscapes resist being influenced by interventions or the

"influence of regime and niche actors" (Geels and Schot 2010, p.19). The MLP approach is one of the "theories of change" that transition design relies on to "inform a deep understanding of the dynamics of change" (Irwin 2015) and "enable designers to effect transitions towards more sustainable systems" (Tonkinwise 2023). Within transition design, strategies incubated at the niche level (e.g., innovative initiatives, actions or cultural movements emerging in the context of existing problems, norms and infrastructure in regimes and landscapes) are often considered "the seeds for transitions" (Geels and Schot 2010, p. 24) and can spread by aligning with the "predictable trajectories" (Geels and Schot 2010, p.21) of the processes and dimensions of the regimes. At the regime level, developments may generate sufficient pressure to influence regimes' trajectories, generating 'windows of opportunity' for novelties with adequate momentum and stability to break in and provoke adjustments. In turn, the new regime may influence the landscape (Figure 1). On the contrary, if the niche initiatives do not alter the elements in the regime, they will remain niche-bound and, eventually, fade. In other words, transition will only occur if the processes at different levels of socio-technical systems "link up and reinforce each other" (Geels and Schot 2010, p.27).

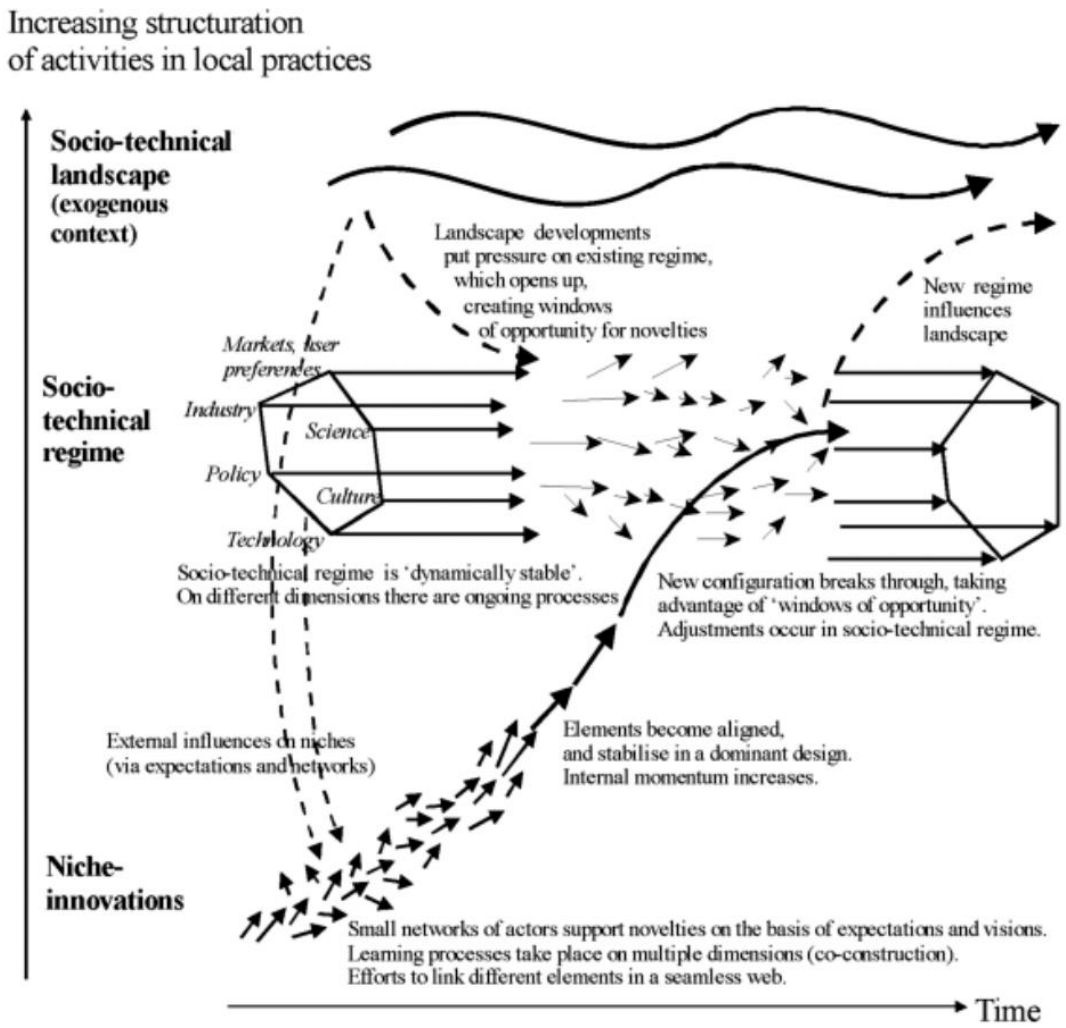



Figure 1. Multi-level perspective on transitions. Source: Geels and Schot 2010 p.25.

While the MLP seems to be a good analytical tool to define and understand the potential for transition in systems, some researchers argue that the framework focuses on compartmentalized broad sectors of society – industry, science, policy, technology, etc. – to improve their efficiency and resource consumption, lacking a closer connection with the complexity of reality and tangible aspects of human experiences (see, for example, Kossof 2011, Irwin et al. 2020, Wallace 2021). Alternatively, Kossoff's (2008, 2011) Domains of the Everyday Framework proposes to apply the multi-level perspective while adopting a 'whole system' approach, conceiving social forms – households, neighbourhoods, cities, regions and the planet – as semi-autonomous and

mutually interdependent ‘wholes’ working together for greater missions. For integrating the MLP approach and the Domains of Everyday framework, Wallace (2021) argues that niche activity, regimes, and landscapes exist in the domains of household, neighbourhood, city, region, etc. and suggests that transition relies on the processes and dimensions of niches, regimes, and landscapes that exist at every scale of the Everyday.

Practices in transitions

As introduced in the previous section, everyday life is essential in transition design theory, and it is strongly connected with practices. Several researchers have explored Social Practice Theory in relation to change in complex systems (see Hargreaves et al. 2012 and 2013, Kossoff et al. 2015, Watson and Shove 2022). Everyday life can be understood as a network of interdependent ‘practices’, that Watson and Shove (2022) defined as “socially shared patterns of activity”. Practice theory (as developed by Giddens 1984, Schatzki 2016, and Shove et al. 2012) is recognised for its value in analysing and understanding ‘large’ and complex social phenomena, such as transitions. According to Shove et al. (2012), people’s and communities’ daily behaviours and experiences determine the fundamental context within which more sustainable directions can be “steered”, so practices can be conceived as the basic unit of transformation. Shove et al. (2012) establish that “practices are defined by interdependent and mutually shaping relations of materials, meanings and competences”. Consequently, a practice can be influenced by altering its elements – that is, the materials, meanings, or competences associated with it – or by altering how such elements integrate (Shove and Watson 2010). A new practice can also emerge by innovatively combining new or existing elements (Shove et al. 2012).

Relationships between practices are also central to establishing effective, long-lasting, and enduring practices (and, ultimately, influence transitions). Shove et al. (2012) studied how practice bundles and complexes form or dissipate due to competition and/or collaboration between practices. For example, they discuss how certain practices are interdependent: for some to be performed, others must be previously performed (e.g., for reading a text, the text has to be written). In most cases, the elements of practices (materials, meanings and competences) are more than just “building blocks”;

they work as aggregators and vehicles of practices, becoming the point of connection among them and playing a significant role in how these connections evolve. Sometimes, links between practices are established not by shared common elements, but by elements that are closely located (Shove et al. 2012).

Like Shove and Watson (2010), transition researchers such as Kuijer (2014) or Irwin et al. (2021) urge designers to (re)design practices performing at different Domains of the Everyday. Transition theory argues that the external satisfaction of needs predominates in the unsustainable models of modern societies, and claims the pursuit of 'authentic holism' – that is, when the 'parts' of a whole conform a self-functioning and semi-autonomous unit – to ultimately achieve ecological, social, economical, political and cultural sustainable transitions (Kossof 2011, Kossoff et al. 2015).

Experimenting with practices as prototyping in Transition Design

The previous sections indicate significant methodological implications in the work within transitions. First, practices should be conceived as the basic unit of design. Second, practices can be altered or established by (re) organising their elements – materials, competences and meanings – or the way those elements integrate. Third, practices influence each other, and can be aggregated into bundles that share common elements. Fourth, practices unfold from the satisfaction of needs, so design should attend to how needs are satisfied. Fifth, the internal satisfaction of needs in the Domains of Everyday may facilitate transitioning to more sustainable models. Sixth, niches, regimes and landscapes operate within the scales of the Domains of Everyday, meaning that the design of practices within a Domain can transform such Domain and influence practices in other Domains. Taking these considerations into account, we understand practices as the main object of design for transitioning towards sustainable systems and, consequently, as the object of prototyping in such projects.

Aligning with researchers studying complex systems, we hypothesise that prototyping can play a significant role in addressing transitions (Björgvinsson et al. 2010, Kimbell and Bailey 2017, Bijl-Brouwer and Malcolm 2020, Drew et al. 2021). Yet, we acknowledge that the practical and methodological implications of the application of prototyping in complex transformations remain underexplored (Maiorana 2021, Drew et al. 2021). In the study of prototyping within teams approaching systems change, Bijl-Brouwer and

Malcolm (2021) and Vitaller et al. (forthcoming) provide insights on the application of an evolutionary approach and a portfolio of interconnected prototypes that support the understanding of the envisioned system change. Maiorana (2021) offers an overview of the characteristics of complex systems as a starting point for defining the features of prototypes operating within complexity. Both socio-technical and Social Practice Theory consider that, given the impossibility of anticipating the effect of intervening in a system, interventions should be approached reflectively and iteratively (Irwin et al. 2020). Relatedly, Wallace (2021 p.10) identifies a common approach to complex systems based on "safe-fail" temporary experiments that provide "small-scale ways forward to avoid catastrophic impact" that may also inform about existing tensions between niches and regime norms when working on transitions. Specific research on prototyping practices within transitions or systemic change is also scarce. Some studies have stressed the value of prototypes as socio-material configurations embedding the performance of practices (Suchman et al. 2002, Danholt 2005, Wilkie 2010), that can already exist or be entirely new (Ehn 1988, Mogensen 1992). This idea connects with the breaching experiments run by Garfinkel (1967, p.57), designed to deliberately disrupt the expected social norms and regular uses of public space to examine people's reactions. Prototyping practices could, to some extent, provoke by suggesting new unexpected (even disrupting) social norms and performances, as a means to constructively promote and establish new configurations, behaviours and social order (Wilkie 2010 p.43). Despite this, there is a gap in the methodological and practical development of prototypes targeting practices that could significantly influence the work in transition design processes. This paper aims to contribute to the closure of this gap by addressing the following research question: How can the prototyping of practices support design processes aiming for sustainable transitions?

## Methodology

This paper builds on the case study of the pilot in Jerez de la Frontera of PALIMPSEST, a practice-oriented research project funded by the European Commission between 2023 and 2026. The project investigates the contribution of artistic and cultural participatory initiatives in transforming landscapes facing climate challenges. Our case study aims to support theory construction grounded in empirical evidence (Eisenhardt and Graebner,

2007) and offer insights through a detailed examination of a real-world context (Yin, 2009). The data, derived from the prototyping activities undertaken by the Jerez team, identifies methodological and theoretical insights that can be applied to similar cases. Three co-authors were directly involved in the project consortium and closely followed the planning and prototyping activities. The data was collected over two years, through participant observations of prototyping activities, workshops and planning meetings (Czarniawska 2012), and archival analysis (Mills & Mills 2018) of the project's prototyping documentation (Figure 2). The arguments in this paper are distilled from a thematic analysis (Braun and Clarke, 2006) of the various data sources (Table 1).

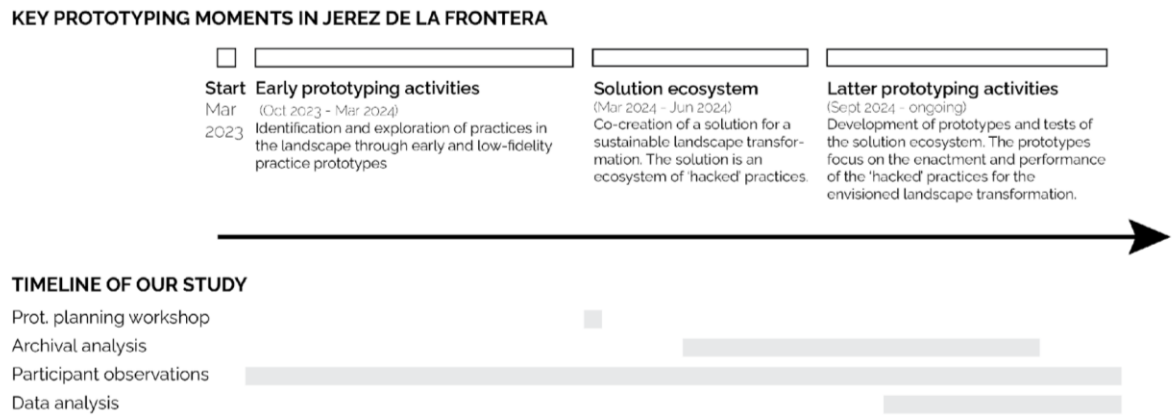



Figure 2. Timeline of data collection and analysis activities. Source: authors

| DATA COLLECTION METHOD | DESCRIPTION |
|---|---|
| Prototyping planning workshop | The prototyping planning workshop in Jerez involved 10 members of the consortium, including the local team. It facilitated the discussion of the prototyping approach in PALIMPSEST and initiated the work on the prototyping plan. Notes from direct observations and relevant quotations were arranged according to initial themes and contributors. The main outcome of the workshop - the draft of Jerez's prototyping plan - |

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|---|---|
| | was analysed to establish initial codes that guided the tables used in the archival analysis. |
| Archival analysis | The primary source materials for the archival analysis include six working documents related to the prototyping planning and reporting, including: a narrative document of the envisioned solution, a Gantt chart, a budget planning spreadsheet, and three project deliverables and reports. The blog written by the Jerez team to disseminate project activities to local stakeholders also served as a rich data source. We conducted an in-depth content analysis of these documents and coded and organised the information about each prototype in a spreadsheet. For each prototype, we coded information about its filtering dimensions (e.g., goals and focus, the targeted aspect of the original or 'hacked' practices to be evaluated), manifestation dimensions (e.g., materialisation techniques, fidelity in relation to the envisioned solution), and the participants and methods used for testing. The entire list is shown in Annex 1, along with an example of one of the prototypes. |
| Participant observation (from March 2023 to April 2025) | The transcriptions, video recordings, and notes from the PALIMPSEST biweekly prototyping planning meetings were arranged in sequence and summarised into abstracts after each meeting, including relevant captions (date and contributor). Then, the materials were annotated following the codes established for the archival analysis to complete the information in the tables. A loose thematic analysis supported the understanding of some of the decisions and challenges faced by the team in the prototyping process. |

Table 1. Data collection

Relevance of the case study

PALIMPSEST investigates ways to transition to more sustainable landscapes through the activation of cultural and creative initiatives that respond to challenges affecting multiple systems. To do so, the project activated a process at three pilot locations – Jerez de la Frontera (Spain), Lodz (Poland) and Milan (Italy). Figure 3, retrieved from the project proposal, visually represents such processes (an extensive description of PALIMPSEST approach to landscape transitions is provided in Bianchi et al. 2024) and shows how PALIMPSEST is inherently connected to prototyping. Not only is prototyping an essential stage of the proposed process; the project proposal explicitly describes PALIMPSEST as an "experiment" on the role and potential of creative and cultural initiatives in the transition to more sustainable landscapes (PALIMPSEST 2020). This centrality given to prototyping in the context of landscape transitions positioned PALIMPSEST as a relevant case for our study.

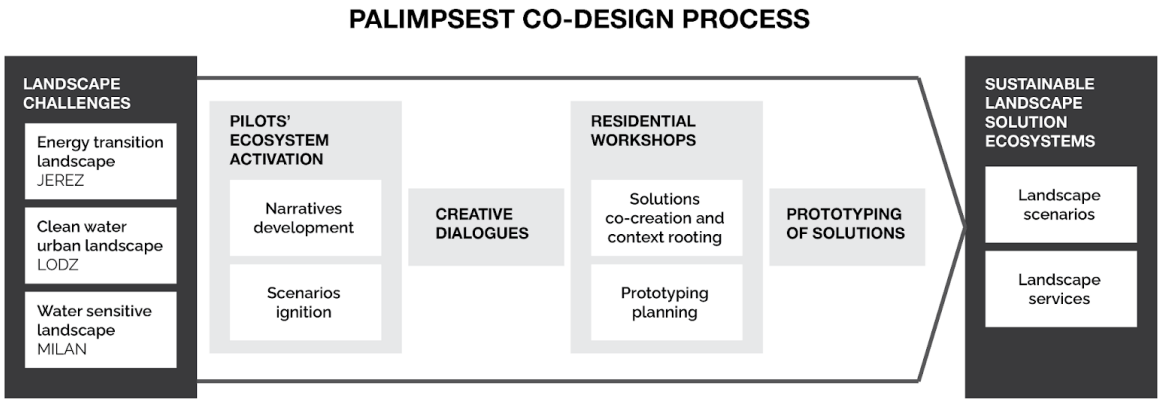



Figure 3. Timeline of the process activated at PALIMPSEST pilot locations to facilitate landscape transitions. Source: authors

Among the three pilots, we decided to focus our study on the case of Jerez de la Frontera because of its distinctive approach to prototyping practices. The team proposed an ecosystem solution that leveraged long-established and deeply rooted practices of the territory to facilitate the transition to more sustainable landscapes. While a similar process was observed in the other two pilots, the prototyping process

followed by the team in Jerez exemplifies important methodological considerations that can illustrate the complex work of prototyping practices in processes targeting sustainable transformation. Being the first one initiating the prototyping activities, the pilot served as a reference for the other two. Particularly, we focused our study on how prototyping existing and new practices unfolded, and the distinctive features of the filtering and manifesting dimensions of the prototypes throughout the process.

## Findings

From existing to hacked practices

Jerez's prototyping plan illustrates that the object of their prototyping activities were the practices in the solution ecosystem that they were aiming to implement in the Jerez context. Starting from existing practices, the team envisioned a new ecosystem of practices to support the transformation of the local landscape into a more sustainable one. The new solution ecosystem was based on leveraging three main groups of practices: 1) the fabrication of Zambomba [1]instruments, 2) the performance of the Zambomba festival and 3) the renaturalisation of Jerez's urban landscape (Figure 4).

[1] The Spanish term 'zambomba' refers to two things: a percussion drum-like instrument, and a participatory tradition where the instrument is played to accompany carols and other popular songs around a firepit. The instrument consists of a ceramic vase closed with a piece of leather crossed with a wood rod. By rubbing the rod with both hands, the vibration produces zambombas' particular sound. Zambombas are inherent to Jerez's culture, and are experiencing a significant comeback in recent years (El Pais 2017).




Figure 4: The solution ecosystem in Jerez leveraged three main groups of practices. Source: authors

The analysis of the targeted practices revealed some commonalities that influenced the design and execution of the prototyping plan. As shown in Table 3, all three groups of practices were already enacted and connected to the landscape in Jerez de la Frontera. Despite the practices of instrument fabrication and performance of Zambomba being linked, the fabrication process of the ceramic basins was outsourced, and only the final assemblage occurred in the local context. There was no connection between the zambomba practices and the renaturalization of the urban space. In addition, the Zambomba festival and the renaturalization practices (in private spaces of wineries in the city centre) had been strongly oriented towards the tourism and recreational industry. Table 2 presents the practices related to the solution already existing in the context, describing its connection with the context, and the associated materials, competences and meanings.

| PRACTICE | CONNECTION WITH THE LOCAL LANDSCAPE | MATERIALS | COMPETENCES | MEANINGS |
|---|---|---|---|---|
| Zambomba fabrication | The ceramic part of Zambomba instruments are developed outside of the Jerez region. Only one family in Jerez (father and son) mantain the assemblage of all the materials into instruments | Ceramic vase, cotton fabric, hardwood stick | Zambomba assemblage, percusion performance, | The family in Jerez that continues assembling Zambombas does it to mantain the legacy, and consider it as a “responsibility” towards their region |
| Zambomba performance | Traditionally, Zambomba rituals occured in private family patios, where (mostly women) gathered to sign traditional songs with lyrics of protest or important lessons to younger generations. In recent years, Zambombas have become a full festival, receive thousands of | Zambomba instruments, firepit, lyric books, | Zambomba performance, singing and dancing, | Local gatherings around a fire to share songs that transmit local knowledge and stories, and cultural heritage. In Zambombas, everyone (kids, elderly, locals or outsiders) are invited to join the circle and sing along, and |

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| | tourists and are more disconnected from its original cultural roots | | | get in the centre to dance |
| Renaturalization of the urban space | Some years ago, one of the local wineries installed a variety of vine in one of their private streets to shadow it. The practice resulted in a smart and aesthetic solution for reducing the heat at the street level. Some of the winery owners, through the local association Amigos de los Árboles, has been trying to promote this practice in the municipality to replicate in public areas, unsuccessfully. | Specific variety of vine, metal wires (structure) | Vine growing, urban planning | The covered alley has become an icon of the winery, attracting tourists and locals for the pleasant and aesthetic atmosphere the vines provide in the hot season. |

Table 2 - Practices leveraged by the Jerez solution ecosystem

Looking at the filtering dimensions of the initial prototyping interventions in Jerez, we can find three groups of goals: a) comprehending the existing practices and their relevance in the context, b) identifying the practices' elements and the actors performing such practices, and c) co-envisioning, together with those actors, possibilities of reformulation of the practices towards a more sustainable landscape. Analysing the materialising dimension, those initial interventions consisted of visits to sites, walk & talks demonstrations of practice, small tests, etc. These prototypes were not traditional experiments to validate a hypothesis or present a specific solution to potential users. Instead, the early prototypes were shaped to collect ideas and understanding of the context by being immersed in the landscape, in touch with the materials, and in contact with the stakeholders involved in enacting the targeted practices. For example, the team organised several visits to the local school of arts in Jerez. In these meetings, they discussed with the teachers the possibility and demands of altering the curriculum in ceramics, presented the idea of the local fabrication of zambombas, visited the facilities, familiarised themselves with the tools and materials employed, ran some tests using the ashes of vine pruning as ceramic glazes, and co-created the curriculum of a ceramic course.

The particularity of the filtering and manifesting dimensions of Jerez's early prototypes prompted important conceptual discussions on the ontology of prototyping and testing in the consortium. The notes and transcriptions of the prototyping planning meetings reveal two critical questions: "What do we conceive as a prototype?" (and, relatedly, "Should we consider this a prototype?") and "What is the difference between prototyping and testing?". The discussions led the consortium to understand such early interventions as prototypes for two main reasons: 1) their intention (i.e., exploring the practices) and 2) the approach, through their materialisation. That is, the team planned initiatives that facilitated the practices' enactment and performance so that the practices could be examined. The latter question, on the difference between prototyping and testing, was connected to the project framework, which established prototyping and testing as separate stages. The meeting notes indicate that consortium members used the two terms indistinctly, and a specific slot of a plenary meeting was dedicated to the topic. The consortium defined prototyping as "the process of generating something (an early model, an initiative, a sample of product or service…) to

explore a hypothesis, demonstrate functionality, or validate an idea". Then, testing was understood as "the process of evaluating the functionality, usability, and/or compliance with expectations or requirements of a product, service or a component, aiming to identify defects, weaknesses, or issues to iterate on". Yet, the team agreed that the characteristics of the solutions developed in the pilots provoked an overlap of the two processes. Jerez's team involved the enactors of targeted practices in developing prototypes. Those enactors, during the prototypes development, evaluated aspects of the idea and identified defects, possible opportunities, or necessary iterations during the prototyping process. Consequently, the prototype was iterated before the testing. On other occasions, the prototyping required the consultation of decision-makers or technical experts (e.g., ceramists) who would not necessarily interact with a complete prototype. Nonetheless, they would provide feedback and relevant insights to develop the prototype to be tested by a broader audience. Returning to the example of the ceramic course in Jerez, before testing the new format of the ceramic course, the team, together with the artist and the teachers of the program, prototyped some of the sessions: they went to visit the rural landscape and considered the materials to use as clay and as enamels for the zambomba instruments, run some firing tests with those materials, shaped zambombas as examples, etc. During these activities, labelled as "prototyping" in the pilot plans, numerous evaluation moments occurred, allowing them to make decisions for the later prototype of the program tested with students between September and December 2024.

## Prototyping hacked practices in socio-technical systems

The results of the initial prototyping interventions informed the further development of the solution ecosystem. Thanks to the understanding of the materials, spaces, and stakeholders involved in the practices, the Jerez team had a clearer idea of how to 'hack' such practices for the envisioned sustainable transformation of the landscape. Following our example, they detailed a plan to alter the existing educational program on ceramics as a way to transform the way that local students relate to the materials and the cultural heritage of the landscape. Also, the knowledge obtained about each group of practices facilitated the combination and interconnection among the targeted

practices to promote sustainability. To do so, one of the strategies involved 'hacking' the principal object of Zambomba rituals – the instrument – as a connector among the three blocks of practices. The solution conceived the Zambomba instrument as an object in transformation, changing in function and shape, depending on the practice enacted at a particular moment (Figure 5). First, the zambomba would promote the performance of ceramic training and caring practices in relation to the landscape. Then, the instrument would be played at the Zambomba festival. While this is a regular practice, the material components of the instrument, coming from the local landscape (e.g. clay from the traditional albariza soil of the area, enamels from pruned vegetal leaves and branches, etc.), would alter the meanings associated to the performance, physically bringing the rural landscape stories and heritage in to the urban context. Last, by the end of the festival, the instrument would be broken at the bottom, transforming into a pot. The instruments converted into pots would host a specific type of vine in strategic areas of the city centre and establish a natural cooling system in the city. The pots would be planted and maintained by local business owners, promoting a chain of caring practices between residents and the rural landscape that also thrives (and provides care) in urban areas.

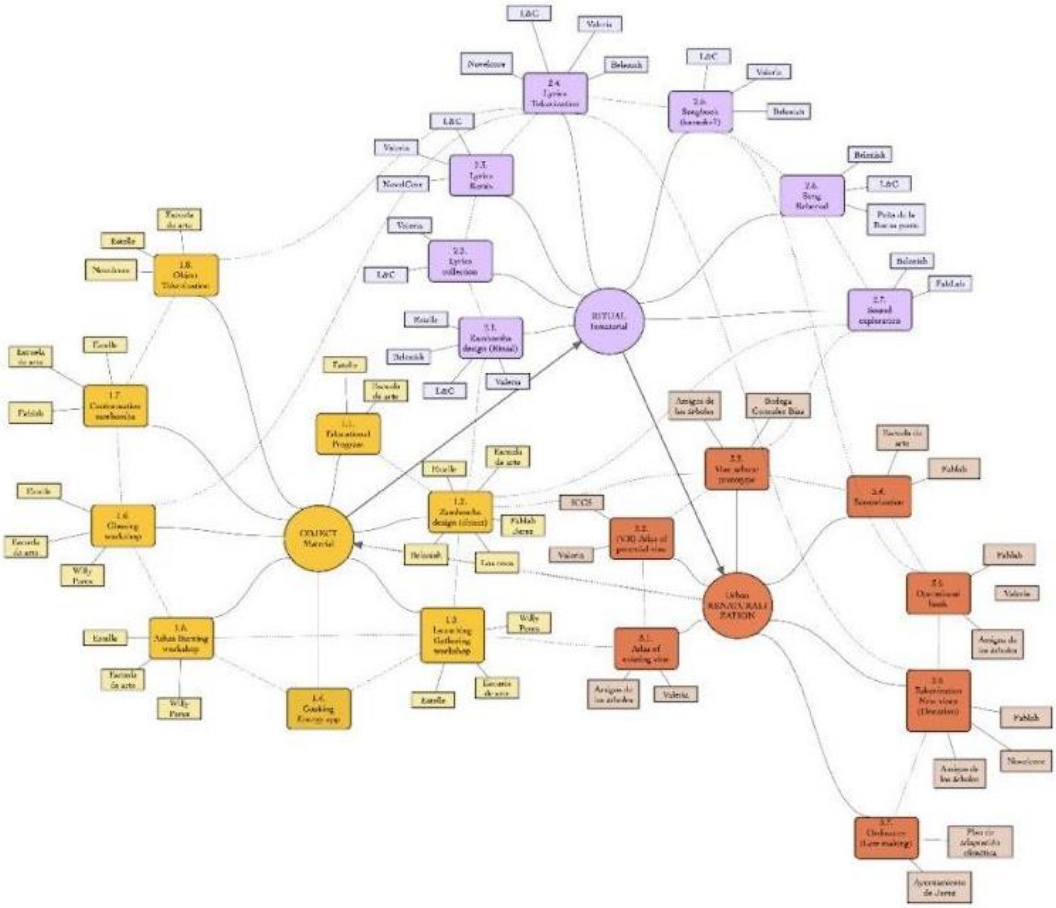



Figure 5. The cycle of the Zambomba object in Jerez's solution ecosystem. Source: PALIMPSEST Jerez team

Some of the practices were 'hacked' not only by altering the materials, but also part of the meaning (e.g. the lyrics of the songs were altered to trigger conversations about the implementation of renewable energy plants in Jerez's rural landscape) or the competences (e.g. the fabrication of Zambombas in the ceramic program also incorporates competences connected to the landscape conservation and cultural heritage). Furthermore, there were alterations in the physical location where practices are enacted (especially in the case of the instrument fabrication, fully developed in Jerez), and the interconnection between practices that were previously disconnected

(like the zambomba making and the ceramic crafts programme at the School of Arts). All those alterations attempted to change the performance of the practices and, thus, provoke a transformation of the system towards sustainability.

The filtering and manifesting dimensions of the latter prototyping activities in Jerez, oriented to evaluate the performance and effect of the 'hacked' practices in the context, significantly differ from the early ones. In the goals' descriptions (within the filtering dimension) we appreciate a shift from the use of verbs like "understanding" and "exploring" – predominant in the early prototyping – to "evaluating", "assessing", or "testing". In general, the goals of the prototypes could be classified according to four primary objects of inquiry: 1) the elements of the practices and their (re)integration in the proposed practices, 2) the enactment of the 'hacked' version of the practices by the stakeholders in the local context, 3) the integration among groups of practices to institute the solution ecosystem and 4) the required (adjustment of) rules and norms associated with the proposed practices (see Table 3). Yet, most of the latter prototypes targeted more than one of the four focuses. Prototypes addressing focus 1 and 2 were more numerous, and sometimes several were tested as part of the same prototyping intervention. Prototypes addressing focus 3 and 4, given the scale, demanded more time and resources in the preparation and execution and were, therefore, less frequent.

| IDENTIFIED PROTOTYPING FOCUS | EXAMPLE OF A PROTOTYPE |
|---|---|
| 1. Elements of the practices and their (re)integration in the proposed practices | A series of firing tests using different enamels from pruned vegetal leaves and branches to evaluate their behaviour on the clay that would be used for the zambomba making (April-May 2024) |
| 2. Enactment of the 'hacked' version of the | Pilot kick-off of the ceramic crafts program (September 2024) focusing on the Zambomba, including a) dialogues |

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| practices by stakeholders in the local context | and presentations with experts on the instrument, b) walks and material collection in the rural landscape for student to engage with a bioregional approach and 3) visit to the archeological museum, focused on the traditional vases and containers from the area that inspired the Zambomba instrument |
| 3. Integration among groups of practices to install the solution ecosystem | During the firing of the Zambomba instruments at the School of Arts (December 2024), the fires were also used as a reference to the Zambomba ritual. Coinciding with the dates of the zambomba festival in the city, the PALIMPSEST team members displayed some of the elements present in Zambomba rituals (musicians, traditional lyric books, popular food) around the firepit and invited the students and teachers to participate in an rudimentary test of the gathering and singing performance. |
| 4. Required (adjustment of) rules and norms associated with the proposed practices | The Jerez team has initiated and facilitated a series of regular meetings and dialogues with the Municipality of Jerez where they presented, using different formats (photomontage, slidedeck, visits to the site, testimonials) the idea of the renaturalization of certain areas of the city centre and discussed with the municipality possible ways to facilitate the implementation of the idea by changing the current norms that prevent private groups to develop green infrastructure initiatives in public spaces. |

Table 3 - List of focuses identified in the latter prototyping in Jerez

Our analysis also showed a clear difference in the manner that the latter prototypes were manifested, in terms of the techniques employed and the fidelity levels. The latter prototypes attempted to perform the proposed practices in scenarios as close to reality

as possible so that the participants could enact the envisioned transformation in the context. This resulted in high-fidelity pilot tests. Relatedly, the budget and other resources available for the prototyping activities were questioned. Even those prototypes that filtered the evaluation of elements of the proposed practices, or their enactment were developed in high fidelity in respect to the envisioned solution (Table 3). This approach seemed to respond to different concerns related to the prototyping and testing stages, voiced during project meetings. First, the presence of decision-makers and influential stakeholders interacting with the test triggered the team to adopt a higher level of formality and realism when developing the prototypes. If the new practices were to become the norm, their prototypes should provide a realistic experience for key stakeholders to conceive what would be needed to integrate them into the existing landscape of rules and regulations. Second, to strengthen the credibility of the interventions in the context for participants, so that they naturally react to the proposed practices and begin to enact the envisioned landscape. Lastly, the prototypes were developed in collaboration with local craft makers, artists, and other professionals, so they became a tangible portfolio that showcased their skills and capabilities. The prototypes materialised as in-context performances, activities, workshops, or piloting interventions, where representative stakeholders (citizens, local associations, landowners, business owners, students, teachers, technical experts, and decision-makers) enacted and/or experienced the proposed practices. The pilot team stressed the importance of guaranteeing the simultaneous participation of the actors involved in the practices to ensure two crucial elements in prototyping the complex transformation: 1) the accuracy and reliability of the results indicating the effect of the proposed 'hacked' practices, and 2) the emergence of discussions about the proposed visions from multiple perspectives and opinions.

## Discussion

Our findings present relevant insights regarding the prototyping of practices as a strategy for sustainable transitions. The Jerez team conceptualised and prototyped a solution that leveraged existing practices to promote the envisioned change in the

landscape. These practices were initially separated, unrelated to each other, and in some cases, also external to the local context, such as Zambomba production. The team used a series of prototypes to better comprehend the elements and characteristics of the practices' performance in the context, as a first step in understanding their transformational potential. This resulted in an ecosystem solution that 'hacked' the practices. The team shuffled practices that were deeply rooted in the context by aggregating them (Shove and Watson 2010), centralising their performance (contributing to making them more 'endogenous' (Kossoff 2011, Kossoff et al. 2015)) and/or editing the elements conforming them.Then, they developed a number of prototypes to evaluate the adoption and enactment of the 'hacked' practices in context, as well as their impact in relation to the envisioned transformation. These results resonate with existing literature on practice theory and transition design theory claiming the potential of practices as an essential unit of design for transitions (Shove and Walker 2010, Wallace 2021).

Our case shows a good example of different aspects that may support 'hacking' practices for envisioned transitions. Designers may alter their elements – materials, competences, meanings – , the connection (or separation) between practices, or their location (and/or actors) of performance.  Still, the team of Jerez stressed the value of balancing the 'hacked' and remaining aspects of the original practices, so that the familiarity eases participants' engagement and enactment of the proposed version. This resonates with Ehn's "dialectics of tradition and transcendence" (Ehn 1988, p.129); the dilemma of developing prototypes that support existing practices or establishing unconventional ones (Mogensen 1992, p. 1).

These results also extend the literature of prototyping in design processes targeting complex transformations, contributing to close the existing gap on the topic. The case of Jerez reflects clear differences on the prototyping approach applied throughout the process, leveraging what Floyd (1984) termed as "exploratory" prototyping approach for the earlier prototypes, and the "experimental" approach in the latter ones. While this sequence has been previously found in more traditional design processes (Kimbell & Bailey 2017), it seems to support complex transformations too. The team's reflections about the development of high-fidelity prototypes so that they can serve as a first step

to establish their proposed practices in the long term also hint at an evolutionary approach (Floyd 1984, Bijl-Brouer and Malcolm 2020). By materializing the 'hacked' practices, they become part of reality; the prototypes are constructed in an attempt to train local stakeholders on how to enact for a more sustainable future of the local landscape, situating the possible future into the present. As such, prototypes can almost be conceived as potential niches at different scales of the city (Wallace 2021), with the hope that they will gain traction and eventually influence the socio-technical regime and landscape of the city.

The study of the filtering and manifestation dimensions (Lim et al. 2008) of Jerez's prototypes also reveal insightful methodological learnings. Targeting practices as the unit of design, the filtering dimension of the prototypes in Jerez was always connected to them. Either to understand the targeted practices and their characteristics of performance, or to evaluate the functioning of the 'hacked' practices, the team focused on (or 'filtered', in the terms of Lim et al. 2008), the leveraged practices. This finding suggests that practices (and more in particular, practices' elements, the possible aggregation of practices, the location and actors enacting the practices) could be the guiding principle of the filtering dimension of prototyping in transformation processes. Studying the manifestation dimension of the prototypes, we observed significant differences between the earlier and later prototypes of the Jerez team. The differences in the prototypes' manifestation aligned with the shift in focus (and, hence, approach) for prototyping. The early (exploratory) prototypes were shaped as spontaneous, semi-guided, and semi-planned actions, with lower fidelity. They involved smaller groups of stakeholders enacting the original targeted practices, mainly in the context of the practices themselves. The later (experimental) prototypes took the form of high-fidelity interventions that generally involved several groups of relevant actors in the traditional context of the practices, required larger amounts of resources dedicated to infrastructure, and simulated the proposed 'hacked' practices at a high fidelity level. The predominance of high-fidelity prototypes poses a critical reflection on the possible consequences of prototyping practices in high-fidelity. On one hand, as the Jerez team stressed, they may facilitate the establishment of prototypes as niche initiatives in the local landscape, and support the collaboration of key stakeholders. On the other hand, previous studies (Niinikoski, 2017) indicate that high-fidelity prototypes may hinder the

emergence of participants' opinions and suggestions for change, as they may be perceived as something 'finished' and unchangeable. This aspect generates crucial methodological questions for practitioners prototyping practices in complex transformation projects.

## Concluding remarks

This paper aimed to investigate how to prototype practices to facilitate sustainable transitions. Drawing from practice theory and transition theory, our research grounds on the idea of practices as the basic unit of transformation. The main contribution of our study are the methodological and practical implications derived from the case study of the prototyping process implemented in Jerez de la Frontera in the PALIMPSEST project. Our paper describes the filtering and manifesting dimensions of the prototypes they developed to establish a solution ecosystem of 'hacked' practices to facilitate the transition to a more sustainable local landscape.

Along with the methodological implications, our investigation also shows valuable insights on the contribution of prototyping to the application of transition theories. The solution proposed in Jerez was based on 'hacked' practices. Those 'hacked' versions of long-standing practices were bundled, more endogenous, and operated at different scales of the everyday. Yet, they maintained some links to the original practices, easing participants' adoption and enactment of the proposed practices and, consequently, contributing to establish them as 'niches' that could eventually consolidate at the regime and landscape levels. We acknowledge our study is only based in one case and context, but we believe further research in similar projects could significantly benefit the work in transitions, digging into the specific mechanisms to consolidate prototypes into regime and landscape levels for the transformation of socio-technical systems.

## References

Bijl-Brouwer, M., and Malcolm, B. (2020). Systemic Design Principles in Social Innovation: A Study of Expert Practices and Design Rationales. She Ji: The Journal of Design, Economics, and Innovation, 6(3): 386–407. https://doi.org/10.1016/j.sheji.2020.06.001

Björgvinsson, E., Ehn, P., & Hillgren, P.-A. (2010). Participatory design and "democratizing innovation." Proceedings of the 11th Biennial Participatory Design Conference, 41–50. https://doi.org/10.1145/1900441.1900448

Braun, V., and Clarke, V. (2006). Using Thematic Analysis in Psychology. Qualitative Research in Psychology 3(2): 77–101.

Brown, T. (2009). Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation. United States: HarperCollins.

Czarniawska, B. (2012). Organization Theory Meets Anthropology: A Story of an Encounter. Journal of Business Anthropology 1: 118–140.

Danholt, P. (2005). Prototypes as Performative. Aarhus, Denmark: ACM Press.

Design Council and The Point People. (2021). System-shifting Design: an Emerging Practice Explored.

Drew, C., Robinson, C. and Winhall, J. (2021). How is design evolving to create the systems we need? An emerging practice of system-shifting design. Design Council & The Point People.

Ehn, P. (1988). Work-Oriented Design of Computer Artifacts. Arbetslivscentrum, Stockholm ISBN: 91-86158-45-7

Ehn, P., Nilsson, E.M. and Topgaard, R. (2014). Making futures: Marginal notes on innovation, design, and democracy. The MIT Press.

Eisenhardt, K. M., and Graebner M.E. (2007). Theory Building from Cases: Opportunities and Challenges. Academy of Management Journal 50, 25–32.

El País (2017). In Spain's Jerez de la Frontera, flamenco zambombas make a comeback. Retrieved from El País in English (digital).

https://english.elpais.com/elpais/2017/12/22/inenglish/1513947053_271989.html [January 2025]

Floyd, C. (1984). A systematic look at prototyping. In Approaches to Prototyping. Edited by R. Budde, K. Kuhlenkamp, L. Mathiassen and H. Zullighoven. 1-18. Heidelberg: Springer-Verlag.

Forman, R. T. T. (2014). Urban Ecology : Science of Cities.

Garfinkel, H. (1967). Studies in Ethnomethodology.

Geels, F.W. (2010). Ontologies, socio-technical transitions (to sustainability) and the multi-level perspective, Research Policy, Vol. 39, No. 4, pp.495–510.

Geels, F.W. and Schot, J. (2007) Typology of sociotechnical transition pathways, Research Policy, Vol. 36, No. 3, pp.399–417.

Geels, F.W. and Schot, J. (2010) The dynamics of transitions: a socio-technical perspective, Part 1 in Grin, J., Rotmans, J., and Schot, J., in collaboration with Geels, F., and Loorbach, D. Transitions to Sustainable Development New Directions in the Study of Long Term Transformative Change, Routledge, New York.

Giddens A (1984) The Constitution of Society: Outline of the Theory of Structuration. Oakland, CA: University of California Press.

Hargreaves, T., Longhurst, N. and Seyfang, G. (2012) Understanding Sustainability Innovations: Points of Intersection between the Multi-Level Perspective and Social Practice Theory.

Hargreaves, T., Longhurst, N. and Seyfang, G. (2013) Up, down, round and round: connecting regimes and practices in innovation for sustainability, Environment and Planning A, Vol. 45, No. 2, pp.402–420.

Irwin, T. (2015) Transition Design: A Proposal for a New Area of Design Practice, Study, and Research, Design and Culture, 7:2, 229-246, DOI:10.1080/17547075.2015.1051829

Irwin, T. & Tonkinwise, C. & Kossoff, G. (2020). Transition Design: The Importance of Everyday Life and Lifestyles as a Leverage Point for Sustainability Transitions. Cuadernos del Centro de Estudios de Diseño y Comunicación. 10.18682/cdc.vi105.4189.

Jacobs, J. (1961). The Death and Life of Great American Cities. Vintage

Kimbell, L. and Bailey, J. (2017): Prototyping and the New Spirit of Policy-making. CoDesign, 13(3): 214–226. https://doi.org/10.1080/15710882.2017.1355003

Kossoff, G. (2008) The Reconstitution of the Domains of Everyday Life, Changing the Change Conference, www.changingthechange.org/

Kossoff, G. (2011). Holism and the Reconstitution of Everyday Life: A Framework for Transition to a Sustainable Society. In: Harding, Stephan, ed. Grow Small, Think Beautiful. Edinburgh (UK): Floris Books.

Kossoff, G., Tonkinwise, C. and Irwin, T. (2015) Transition Design: The Importance of Everyday Life and Lifestyles as a Leverage Point for Sustainability Transitions.

Kuijer, S. (2014) Implications of Social Practice Theory for Sustainable Design, TU Delft, Delft University of Technology.

Lim, Y., Stolterman, E. and Tenenberg, J. (2008). The Anatomy of Prototypes. ACM Transactions on Computer-Human Interaction. 15. 1-27. 10.1145/1375761.1375762.

Maiorana, T. (2021). The Failures of Prototyping: A call for a new definition. In: Proceedings of Relating Systems Thinking and Design (RSD10) 2021 Symposium, 2-6 Nov 2021, Delft, The Netherlands.

Meadows, D. (2008). Thinking in Systems: A Primer. White River Junction, VT: Chelsea Green Publishing.

Meadows, D. H., Meadows, D.L. and Randers, J. (1972). The Limits to Growth : A Report for the Club of Rome's Project on the Predicament of Mankind. London: Earth Island.

Mills, A. J., & Helms Mills, J. (2018). Archival research. The SAGE Handbook of qualitative business and management research methods, 2, 32-46.

Mogensen, P. (1992). Towards a Provotyping Approach in Systems Development, Scandinavian Journal of Information Systems, 4, 31-53.

Niinikoski, Johannes (2017). The effect of prototype fidelity on usability evaluation in product development. Aalto University. Finland.

Rotmans, J. and Loorbach, D. (2010). Toward a Better Understanding of Transitions and Their Governance: A Systemic and Reflexive Approach. In: Grin, John, Rotmans, Jan and Schot, Johan. Transitions to Sustainable Development. London (UK): Routledge.

Schatzki TR (2016) Keeping track of large phenomena. Geographische Zeitschrift 104: 4–24.

Shove E, Pantzar M, Watson M (2012) The Dynamics of Social Practice: Everyday Life and How It Changes. London: SAGE.

Shove, E. & Walker, G. (2010). Governing Transitions in the Sustainability of Everyday Life. Research Policy. 39: 471-476.

Suchman, L., Trigg, R., & Blomberg, J. (2002). Working Artefacts: Ethnomethods of the Prototype, British Journal of Sociology, 53(2), 163-179.

Tonkinwise, C. (2023). Some Theories of Change behind and within transition designing in Strategic Thinking, Design and the Theory of Change (pp.270-293). Elgar Online. 10.4337/9781803927718.00027.

Vitaller del Olmo, M.; de Götzen, A.; Morelli, N.; Simeone, L. (in review). Portfolio Prototyping as an Evolutionary Approach in Landscape Transitions. Learnings from the PALIMPSEST Project. Temes de Disseny Special issue: Design Practices Transitioning Towards Embracing Systemic Worldviews.

Wallace, N. (2021) Using the multi-level perspective for problem articulation, leverage point identification, and systems storytelling in design, J. Design Research, Vol. 19, No. 1-3, pp.106-132.

Watson, M., & Shove, E. (2022). How Infrastructures and Practices Shape Each Other: Aggregation, Integration and the Introduction of Gas Central Heating. Sociological Research Online, 28(2), 373-388. https://doi.org/10.1177/13607804211055495 (Original work published 2023)

Wilkie, A. (2010) Prototypes in Design: materializing futures in Prototyping Prototyping Anthropological Research on the Contemporary ARC Studio Season 1, Episode 2 2010

Yin, Robert K. (2009). Case Study Research: Design and Methods. Fourth Edition. Sage, Thousand Oaks.

## Annex A

| CATEGORY | THEME | EXAMPLE: launching workshop of the ceramics educational programme |
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| Filtering dimensions | What elements of the solution ecosystem does the prototype tackle? | - The coordination of the necessary requirements and resources to launch an academic programme as part of the ceramic craft education, based on the development of zambombas from a bioregional approach. The considered resources are both material and human (teachers, creative agent, and other local stakeholders)<br>- The transmision of the values of the programme to the students through the activities (understanding the bioregion, valuing local materials and practices, experimenting with new assemblages, and collaborating with local creators and activists) |

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| | What current practices does the prototype plug into/intervene? | - Ceramic crafting<br>- Ceramic crafts teaching<br>- Zambomba production<br>- Renaturalization<br>- Wine production |
| | What practices does the prototype aim to stimulate/promote? | - Biorregional ceramic crafting (and teaching) from a closer connection with the rural landscape through materials<br>- Conscious zambomba production, through the use of locally sourced materials and local experts<br>- Connection of the ceramic craft with the local culture<br>- Awareness and critical reflection on the ceramic craft in the landscape |
| | What aspects of the practices is the prototyping action tackling? | - Materials: what materials are employed (and how) in the ceramic craft, their sources, and the impact that their employment generate in the landscape<br>- Competences: on ceramic crafts and zambomba making<br>- Meanings: a critical and reflective approach to the craft and the impact it has in the local enviroment |
| | What dimensions of the local challenge does the prototype study/test/validate? | 1.Recovery of the link with the countryside through intangible heritage (cultural and artistic expressions) that reflect a clear link with the landscape by: |

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| | | a) learning about the strategies and manifestations of local culture that celebrate the relationship between city and territory (in this case, by revisiting existing ones like zambomba rituals)<br>b) leveraging existing practices to promote the transmision of knowledge about human and non-human actors (in this case, zambomba crafting, rural landscape, bioregion, or urban renaturalization)<br>2.Promoting coexistence by:<br>a) learning from existing actions in the city from different groups that work towards a more sustainable landscape in Jerez<br>b) fostering circular processes that can be incorporated in existing events (e.g. academic programme) enabling their continuity and incorporation into cultural circuits (e.g. Zambomba festival) |
| Manifesting dimensions | How are the targeted elements of the ecosystem solution realized in some kind of concrete manifestation? | A pilot 3-days kick-off of the program at the school of arts, where end-users and involved stakeholders will experience presentations, dialogues and workshops in full fidelity |
| | What is the level of fidelity of the prototype in relation to the | Very high |

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| | envisioned solution ecosystem? | |
| | How is the prototype related to the envisioned ecosystem? | It represents several interconnected aspects of one of the main blocks of the solution ecosystem (in this case, the "zambomba as an instrument"). |
| Participation dimensions | Who is participating in the prototyping and testing? | - Students and teachers of the ceramic programme at the School of Arts and Design in Jerez<br>- PALIMPSEST local curators and creative agent<br>- Local stakeholders as experts/informants: zambomba craft makers, musicologist, environmental activist group (Amigos de los árboles), winery owner |
| | Why is it important that they are the participants? | The participants are the promoter of the initiative (creative agent and local curator), the envisioned end-users (students) and responsible agents for the long-term sustainability of the idea (school of arts + local stakeholders) |
| | What is the role of the participants? How do they participate? | Students go through the activities planned in the program, that the promoters and other stakeholders facilitate and support |
| Experiment dimension | How is the prototype being tested/evaluated? | 3-days launching workshop of the education programme in the School of Arts and Design in Jerez. The agenda includes presentations, site visits, walking |

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| | | tours, and dialogues related to the landscape, materials and the craft of zambomba making. |
| | What is the overall purpose of prototype/test? (e.g. to validate, to explore, to learn, to test, to investigate…) | To test (the kick-off of) an academic programme that reconsiders the design of a ceramic object - zambomba - linked to the landscape materials and cultural tradition. The goal is connected to the overall goals of the programme:<br>+ understanding of the bioregion of the Jerez countryside,<br>+ identifying and valuing local materials and existing practices<br>+ experimenting with local materials and creating new assemblages<br>+ collaborating with local creators and activists |

Table 4. Example of the data collected about each of the PALIMPSEST prototypes

## Authors

Maria Vitaller del Olmo, PhD fellow, Service Systems Design Lab, Aalborg University, mvdo@create.aau.dk

Amalia de Götzen, Associate Professor, Service Systems Design Lab, Aalborg University, ago@create.aau.dk

Nicola Morelli, Professor, Service Systems Design Lab, Aalborg University, nmor@create.aau.dk

Luca Simeone, Associate Professor, Service Systems Design Lab, Aalborg University, lsi@create.aau.dk